

Score: / 10

Calculus I (MATH 2200)

Spring 2020

Wednesday, 04/15/2020

Quiz 10: Section 4.4, 4.5, 4.6

Name (Print): _____

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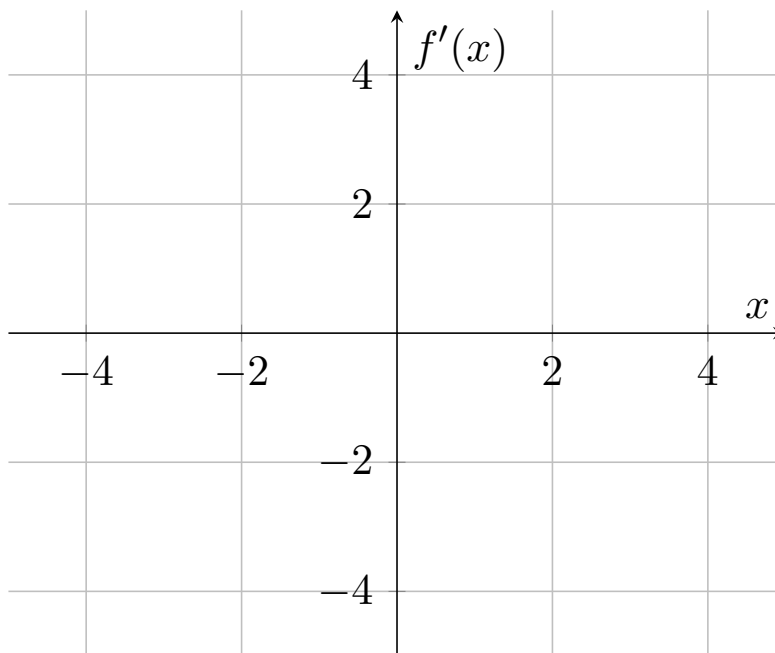
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Section 4.4

3 pts

1. Sketch a possible graph of a function which satisfies

$$f' > 0, \quad f'' < 0 \quad \text{on} \quad (-\infty, 0), \quad f' > 0, \quad f'' > 0 \quad \text{on} \quad (0, \infty)$$



Section 4.5

3 pts

1. Suppose an airline policy states that all baggage must be box-shaped with a sum of length, width, and height not exceeding 64 inches. What are the dimensions a square-based box with the greatest volume under these conditions?
- Ⓐ The box should have a base with sides that are 17 inches in length and a height of 30 inches.
 - Ⓑ The box should have a base with sides that are $64/2$ inches in length and a height of $64/2$ inches.
 - Ⓒ **The box should have a base with sides that are $64/3$ inches in length and a height of $64/3$ inches.**
 - Ⓓ The box should have a base with sides that are 30 inches in length and a height of 2 inches.

Section 4.6

2 pts

1. Find the best linear approximation to $\sin(x)$ at $x = 0$ and use it to approximate $\sin(\pi/72)$.

- Ⓐ The best linear approximation is $L(x) = x^2$ and $\sin(\pi/72) \approx (\pi/72)^2$.
- Ⓑ The best linear approximation is $L(x) = -x$ and $\sin(\pi/72) \approx -\pi/72$.
- Ⓒ The best linear approximation is $L(x) = x + 1$ and $\sin(\pi/72) \approx \pi/72 + 1$.
- Ⓓ **The best linear approximation is $L(x) = x$. So, $\sin(\pi/72) \approx \pi/72$.**

2 pts

2. Find the linear approximation to $f(x) = \sqrt[3]{x}$ at $x = 1$. Is this approximation an overestimate or underestimate? Explain why **without computing the error** of your approximation.

- Ⓐ **The best linear approximation is $L(x) = \frac{1}{3}x + \frac{2}{3}$. Since $f''(1) < 0$, we know this is an overestimate, as the graph of $f(x)$ lies below the graph of $L(x)$**
- Ⓑ The best linear approximation is $L(x) = \frac{1}{3}x + \frac{2}{3}$. Since $f''(1) < 0$, we know this is an underestimate, as the graph of $f(x)$ lies above the graph of $L(x)$
- Ⓒ The best linear approximation is $L(x) = (3x)^{-2/3}$. Since $f'(x) = L(x)$, we know this is an overestimate, as the linear approximation coincides with the derivative.
- Ⓓ The best linear approximation is $L(x) = (3x)^{-2/3}$. Since $f'(x) = L(x)$, we know this is an underestimate, as the linear approximation coincides with the derivative.